

Mid Sem Exam

Sep 2025

Transport Phenomena in Materials Engineering

Max Marks: 30

Duration: 90 mins

- Attempt all the Questions.
- Use of calculator is permitted

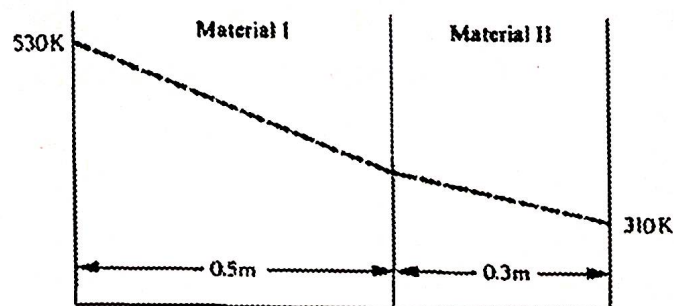
Q1: Derive the expression for generalized Bernoulli's equation for incompressible flow having
(a) Turbulent (b) Laminar nature. (2+3 Marks)

Q2: A viscous molten glass covers a molten metal, and together they flow slowly down an inclined plane that makes an angle θ with the vertical. The thickness of the glass is δ_1 , and the combined thickness of both layers is δ_2 . Each layer has its own viscosity (η_g for glass and η_m for metal). For a plane of length L , assume laminar flow that is fully developed (steady state) and derive an equation for the velocity distribution in each layer. (10 Marks)

Q3: A copper plate of thickness 3 mm is sandwiched between stainless steel plates that are 2mm thick. The left surface of sandwich is held at 400°C and the right surface is held at 100 °C. Given $K_{ss}=17 \text{ W/mK}$, $K_{cu}=373 \text{ W/mK}$. Calculate the temperatures of left and right surfaces of copper. (5 Marks)

Q4: Determine the thermal conductivity of a test panel 150 mm x 150 mm and 12 mm thick, if during a two-hour period $8.4 \times 10^4 \text{ J}$ are conducted through the panel when the two faces are at 290 K and 300 K. (5 Marks)

Q5: At steady state, the temperature profile in a laminated system appears as shown below:



Determine the thermal conductivity of II if the steady-state heat flux is $12.6 \times 10^3 \text{ W/m}^2$ and the conductivity of I is 52 W/mK . (5 Marks)